ProbOS — Month 2, Week 7
FastAPI Service Layer

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Week 7 goal

Expose the kernel over HTTP so it is usable outside a Python REPL – the first step toward external/enterprise-facing usage referenced in `docs/monthly_plans/overall/main.tex`.

Day-by-day plan

Day 1 – FastAPI scaffold

- Set up `python/server/` package structure
- Basic FastAPI app, health check endpoint
- Pydantic schemas mirroring `Model`/`Distribution` constructor signatures

Day 2 – /simulate endpoint

- `POST /simulate`: accept model name, priors, N, `n_steps`; run `MonteCarloEngine`; return percentiles + convergence summary as JSON
- Input validation: sane upper bounds on N/`n_steps` to prevent resource exhaustion (per `docs/standards/qualit` Section 8, deferred item now becoming relevant)

Day 3 – /sensitivity endpoint

- `POST /sensitivity`: run `SobolSensitivity`, return S_1/S_T indices as JSON

Day 4 – /filter endpoint

- `POST /filter`: run `ParticleFilter` against posted observation data, return posterior mean/std trajectory

Day 5 – Integration tests

- `python/tests/test_server.py`: FastAPI `TestClient`-based tests for all endpoints
- Confirm JSON responses match direct Python kernel calls numerically

Day 6 – Pre-week audit + CI integration

- Run `pre_week_audit.sh` before considering the week complete
- Extend CI to install and test the server layer
- Extend `docs/standards/quality_standards.md` Section 8 with the now-relevant API/service checks (previously listed as deferred)

Day 7 – Study guide + documentation

- `docs/study/study_guide.md` entries: FastAPI documentation, Pydantic validation patterns
- Week 7 retrospective appended to this document

Exit criteria

`curl -X POST localhost:8000/simulate ...` returns valid JSON matching a direct Python call to the same `MonteCarloEngine`, for all three endpoints (`/simulate`, `/sensitivity`, `/filter`).